

- VICENTE SCHKOLNIK, *Regressive orders on the non-stationary ideal over a regular uncountable cardinal*.

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Given an uncountable regular cardinal, we introduce several partial orders on the ideal of its non-stationary subsets, as defined in [1], Definition III.6.4. We refer to these as regressive orders. Motivated by Fodor's pressing down lemma (Theorem 8.7, [2]), we show that these orders stratify the non-stationary ideal so that larger sets are closer to being stationary.

We establish that a regressive order makes it possible to regard the smaller of two non-stationary sets as transformable into the larger one via leftward shifts along the ordinal line, while preserving sequences of bounded order type from collapsing into a single point.

We obtain results on the poset structure induced on the ideal by combinations of these orders, including characterizations of maximal elements, antichains, and well-foundedness.

[1] KENNETH KUNEN, *Set Theory*, Studies in Logic Volume 34, College Publications, 2013.

[2] THOMAS JECH, *Set Theory, The Third Millennium Edition, revised and expanded*, Springer Monographs in Mathematics, Springer, 2002.